Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 05)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 compares organic and inorganic fertilisers.

0 1 . 1 Complete Table 1 by giving two advantages and two disadvantages of each type of fertiliser.

[6 marks]

Table 1

Fertiliser	Advantages	Disadvantages
Organic (manure, sewage sludge, compost)		
Inorganic (eg ammonium nitrate, NPK)		

0 1 . 2 Explain how hydroponic cultivation maximises productivity and give **two** disadvantages.

[4 marks]

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0 2 . 1 Explain the effects of deforestation on soil.

[4 marks]

0 2 . 2 Explain how deforestation affects hydrology and climate.

[4 marks]

0 2 . 3 Give **two** ecological impacts of deforestation other than the direct loss of species.

[2 marks]

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0 3 Table 1 compares nitrogen fixation by Rhizobium bacteria in legume root nodules with the Haber process.

Table 1

Feature	Biological fixation	Haber process
Energy source	Sunlight captured by the plant (carbohydrates)	Fossil fuels (natural gas / coal)
Temperature	Soil temperature (10-30 °C)	About 450 °C
Pressure	Atmospheric	200 atmospheres
Catalyst	Nitrogenase enzyme	Iron
Energy density of input	Low	High
Waste products	None - all products used	CO_2, NOx

0 3 . 1 Use Table 1 to explain how natural processes differ from human industrial processes in their use of energy.

[4 marks]

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0 3 . 2 Give **three** other natural processes driven by solar energy that human society depends on.

[3 marks]

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0 3 . 3 Explain how the principle of low-temperature natural processes is being applied in human systems.

[3 marks]

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0 4 Table 1 shows some of the IUCN criteria used to place species in Red List categories.

Table 1

Category	Population decline over 10 years or 3 generations	Area of habitat	Number of mature adults	Probability of extinction in the wild
Vulnerable	over 40%	under 2000 km^2	under 10 000	over 10% in 100 years
Endangered	over 60%	under 500 km^2	under 2500	over 20% in 20 years or 5 generations
Critically endangered	over 80%	under 10 km^2	under 250	over 50% in 10 years or 3 generations

0 4 . 1 State **three** roles of the IUCN.

[3 marks]

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0 4 . 2 A species of lemur has declined from 12 000 to 2 100 mature adults in three generations and occupies 400 km^2 of forest. Use Table 1 to determine its Red List category. Explain your answer.

[3 marks]

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0 4 . 3 Explain why only a small proportion of species have been assessed for the Red List, and give **two** criteria used to decide which species to assess.

[3 marks]

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0 4 . 4 Give **one** example of a species whose Red List category improved because of conservation and state the actions responsible.

[1 mark]

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0 5 . 1 Give **two** wild species with potential as new food species and, for each, one characteristic that makes it useful.

[2 marks]

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0 5 . 2 Explain why indigenous species may be better for domestication than introduced livestock or crops.

[2 marks]

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0 5 . 3 Explain how species of the spinach family could help to solve a problem caused by irrigation.

[1 mark]

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0 6 . 1 State what is meant by **hybrid vigour**.

[1 mark]

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0 6 . 2 Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

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0 6 . 3 Give **two** potential applications of cloning to livestock farming.

[2 marks]

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0 7 . 1 Which of the following would be most suitable for monitoring the movements of a whale over several months? Tick one box.

[1 mark]

A Radio collar ☐

B Acoustic / satellite tag ☐

C Pitfall trap ☐

D Geolocator tag recovered by recapture ☐

0 7 . 2 Explain why post-release monitoring is important in a captive breeding and release programme and describe **two** suitable methods.

[4 marks]

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0 8 A salmon farm rears fish in net cages in a sheltered sea loch. Over one production cycle the farm used 2 400 t of feed and produced 2 000 t of salmon.

0 8 . 1 Calculate the food conversion ratio (mass of feed per unit mass of fish produced) for the farm.

[2 marks]

Answer

0 8 . 2 Explain why so much of the world's aquaculture is of carnivorous species such as salmon, and why this reduces the benefit to world food supplies.

[3 marks]

0 8 . 3 Explain **four** environmental impacts of rearing salmon in net cages in a sea loch.

[4 marks]

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0 8 . 4 Explain **three** ways the environmental impacts of the farm could be reduced.

[3 marks]

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0 8 . 5 Explain why aquaculture cannot simply replace wild capture fisheries.

[3 marks]

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0 9 . 1 Describe how each of the following threatens Antarctica.

[4 marks]

Global climate change .....

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Ozone depletion .....

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Tourism .....

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Future mineral exploitation .....

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0 9 . 2 Describe the main features of the Antarctic Treaty (1959) and the regulations that protect Antarctica.

[6 marks]

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1 0 . 1 State the Brundtland Commission definition of sustainable development.

[1 mark]

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1 0 . 2 Explain the difference between a negative feedback mechanism and a positive feedback mechanism, giving **one** example of each in the climate system.

[4 marks]

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1 0 . 3 Explain why a diverse ecosystem is more likely to resist change than an agroecosystem.

[1 mark]

1 0 . 4 Evaluate the extent to which copying the principles of natural systems could make human activities more sustainable.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

1 1 . 1 Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.

[25 marks]

OR

1 1 . 2 Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS